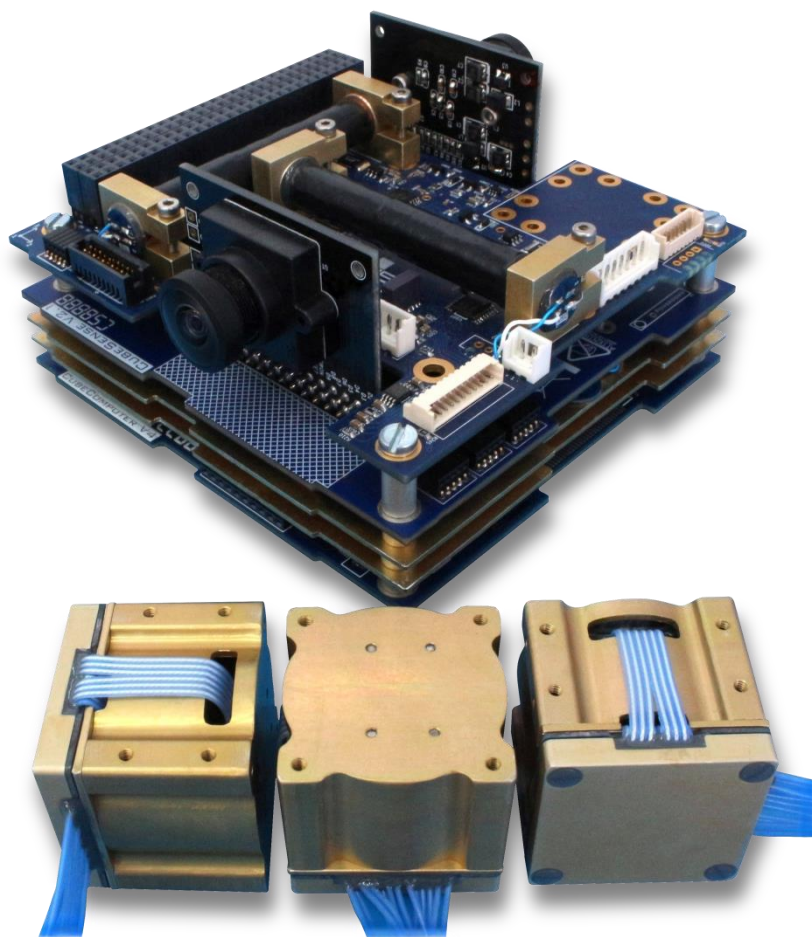




CUBEADCS 3-AXIS

THE COMPLETE ADCS SOLUTION FOR 3-AXIS CONTROL



INTERFACE CONTROL DOCUMENT

ESL
Electronic Systems
Laboratory

Contact Us

Phone : +27 21 808 9499
E-mail : info@cubespace.co.za
Web : www.cubespace.co.za
Facebook : /CubeSpaceADCS
Twitter : @CubeSpace_ADcs

Physical Address

CubeSpace
The LaunchLab
Hammanhand Road
Stellenbosch, 7600
South Africa


UNIVERSITEIT
STELLENBOSCH
UNIVERSITY

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List of Acronyms/Abbreviations

ACP	ADCS Control Program
ADCS	Attitude Determination and Control System
CSS	Coarse Sun Sensor
ESD	Electrostatic Discharge
I ² C	Inter-Integrated Circuit
MCU	Microcontroller Unit
MEMS	Microelectromechanical System
OBC	Onboard Computer
PCB	Printed Circuit Board
RTC	Real-Time Clock
SBC	Satellite Body Coordinate
SPI	Serial Peripheral Interface
TC	Telecommand
TLM	Telemetry
UART	Universal Asynchronous Receiver/Transmitter

1. Introduction

The CubeADCS 3-Axis bundle provides attitude sensing and 3-axis control capabilities to nanosatellites. The unit consists of several CubeSpace modules integrated into a compact bundle. The ADCS onboard computer (OBC) can also serve as the main satellite OBC if necessary and a comprehensive attitude determination and control system (ADCS) software library can be provided with the CubeADCS bundle.

This document will describe the functioning of the CubeADCS unit as well as the mechanical and electrical interfaces to the bundle.

2. Functional Description

2.1 System components

The CubeADCS unit can use combination of magnetometer, coarse sun sensor, fine sun sensor, nadir sensor, MEMS rate sensor, and star tracker measurements to estimate the attitude of the satellite. Magnetorquers and reaction wheels are employed to stabilise and control the satellite's attitude.

The CubeADCS 3-Axis unit consists of up to four integrated PC104-standard PCBs and several peripheral components, which are to be mounted separately. A basic diagram of the complete CubeADCS 3-Axis solution is shown in Figure 1.

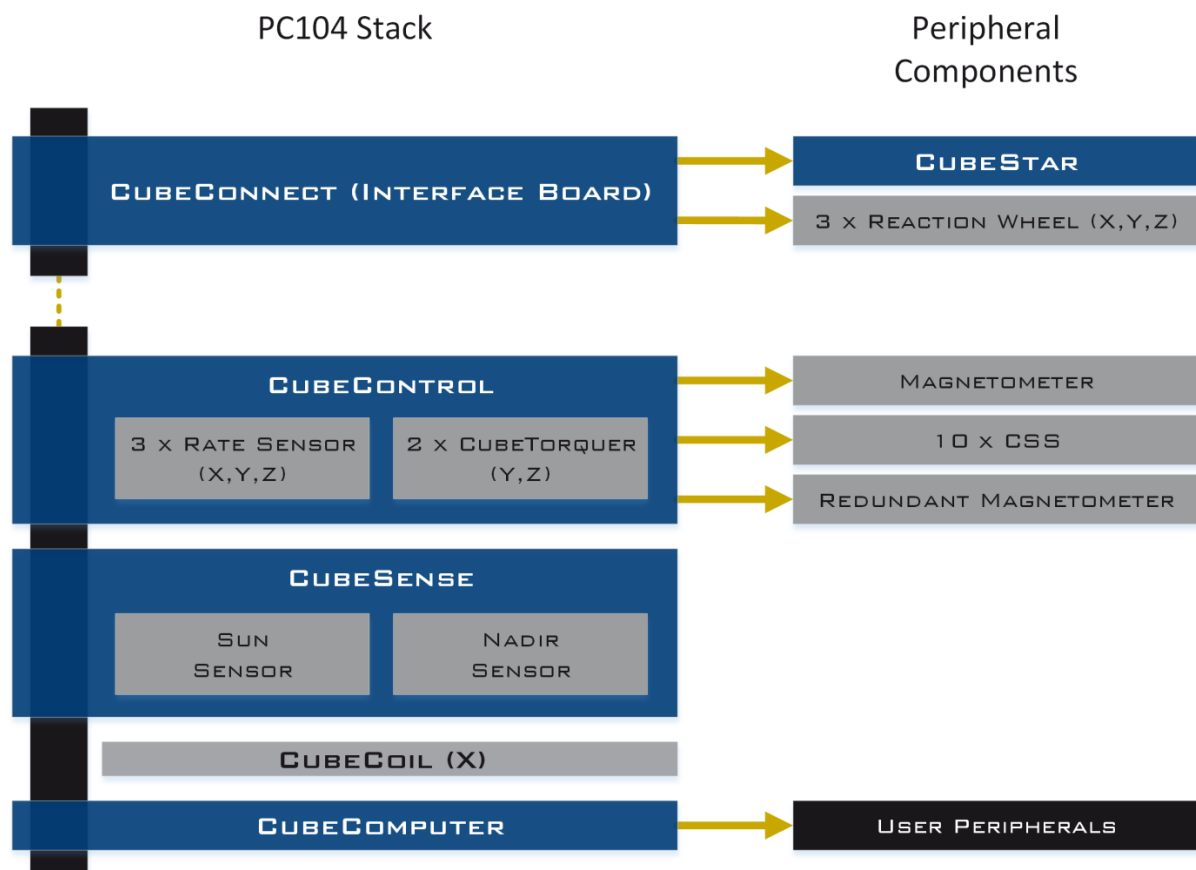
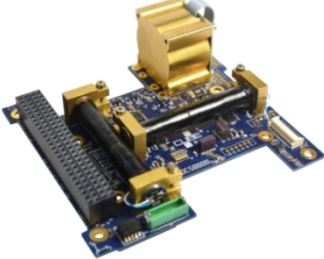
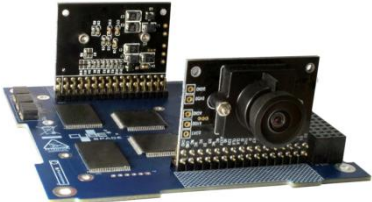
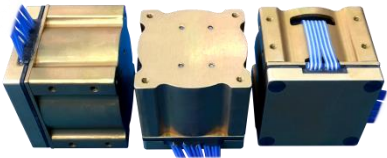


Figure 1 – System diagram of the complete CubeADCS 3-Axis solution

A brief description of the core components of the CubeADCS 3-Axis is given in Table 1.

Table 1 – Modules of the CubeADCS 3-Axis solution

Module	Description
CubeControl 	CubeControl is an actuator and sensor interface module for nanosatellites with advanced attitude control requirements. It is capable of controlling 3 magnetorquers and a momentum wheel. CubeControl can also interface with 2 magnetometers, 10 coarse sun sensors, and 3 MEMS rate sensors. NB: The momentum wheel mounted on CubeControl is not included in the CubeADCS 3-Axis bundle.
CubeSense 	CubeSense is an integrated sun and nadir sensor for attitude sensing. It makes use of two CMOS cameras – one dedicated to sun sensing and another for horizon detection. The sun sensor has a neutral density filter included in the optics. Both cameras have a field of view (FOV) of approximately 160°.
CubeComputer 	CubeComputer is a generic CubeSat OBC. It can perform the required ADCS functions, as well as the satellite's main OBC tasks. The module is based on ARM Cortex-M3 architecture and also implements error detection and correction (EDAC) techniques.
CubeWheel 	CubeWheel is a compact standalone reaction wheel unit for nanosatellites. It provides the satellite with the ability to achieve 3-axis stability and 3-axis control. Each CubeWheel is magnetically shielded and is mountable in 3 axes. Various sizes are available to suit every need.
CubeStar 	CubeStar is a compact star tracker for nanosatellites. The module is based on ARM Cortex-M3 architecture and has significantly low power consumption.

2.2 Body axes definition

The coordinate system definition used by the CubeADCS bundle is shown in Figure 2, whereas the orbit coordinate system is shown in Figure 3. The ADCS coordinate system is furthermore related to the satellite body coordinate (SBC) system through a transformation matrix. When the ADCS is controlling the attitude to zero roll, pitch, and yaw angles, the SBC system will coincide with the orbit coordinate system (referred to as the nominal orientation).

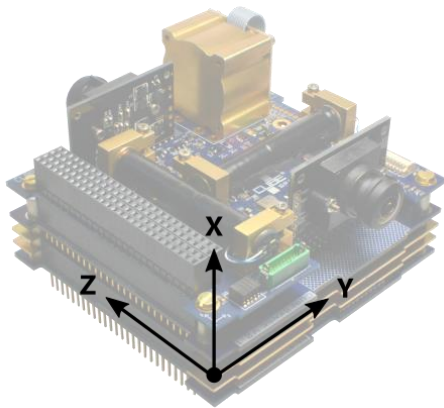


Figure 2 – ADCS coordinate system

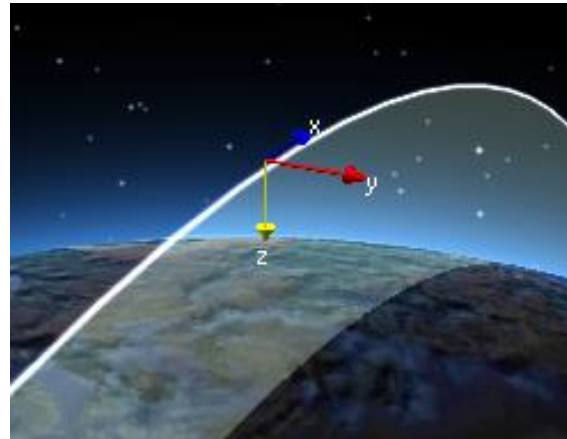


Figure 3 – Orbit coordinate system

2.3 ADCS control loop

The ADCS update function in the ADCS software library must always be executed with the highest priority. The ADCS control loop executes on CubeComputer at a rate of 1 Hz. Sensor measurements and actuator commands are also requested and sent at this frequency.

3. Electrical Interface

3.1 PC104 interface

The CubeADCS unit makes use of the standard PC104 header for electrical interfacing. The pin description of the PC104 bus is shown in Figure 4, as used by the CubeADCS 3-Axis unit.

H2	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	50	52
	1	3	5	7	9	11	13	15	17	19	21	23	25	27	29	31	33	35	37	39	41	43	45	47	49	51
H1	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	50	52
	1	3	5	7	9	11	13	15	17	19	21	23	25	27	29	31	33	35	37	39	41	43	45	47	49	51

PC104 interface pins					
Communication					
H1	1	CANL	CAN bus low		option
H1	3	CANH	CAN bus high		option
H1	21	I2C_SCL_ADCS	Internal I2C clock for all ADCS modules		required
H1	23	I2C_SDA_ADCS	Internal I2C data for all ADCS modules		required
H1	41	I2C_SDA_SYS	System I2C data for CubeComputer		required
H1	43	I2C_SCL_SYS	System I2C clock for CubeComputer		required
H1	17, 18, 19, 20	UART_1	Usable pins for UART_1 (RX or TX)		option
H2	21, 22	UART_1	Usable pins for UART_1 (RX or TX)		option
H1	33, 35, 39, 40	UART_2	Usable pins for UART_2 (RX or TX)		option
H1	29	SPI_CLK	SPI Clock		option
H1	30, 31	SPI_MOSI / MISO	SPI MOSI or MISO		option
H1	32	SPI_CS	SPI Chip Select		option
Power					
H2	29, 30, 32	GND	Ground connection for all modules		required
H2	45, 46	V_Bat	Battery voltage bus		required
H2	25, 26	5V_Main	Main 5 V supply		standard option
H2	27, 28	3V3_Main	Main 3.3 V supply		standard option
H1	47, 49, 51	5V_S	Switched 5 V supply options		option
H1	48, 50, 52	3V3_S	Switched 3.3 V supply options		option
H1	42	BUVIN	CubeComputer backup power		option
Internal ADCS pins					
H1	2, 4, 6, 8	ENABLE	Enable lines for CubeADCS modules		standard option
H2	17, 18, 19, 20	ENABLE	Enable lines for CubeADCS modules		option
H1	5, 7, 9, 11	GPIO / ENABLE	Enable lines for Cubewheels / GPIO		standard option
H1	13, 14, 15, 16	GPIO / ENABLE	Enable lines for Cubewheels / GPIO / ADC		option

Figure 4 – PC104 bus pin description

3.2 Power

3.2.1 Power supply

The ADCS should be supplied with 3.3 V, 5 V, and the raw battery voltage, V_{battery} . The 3.3 V and 5 V buses must be supplied by any one of H1-48,50,52 or H2-27,28 (3.3 V) and any one of H1-47,49,51 or H2-25,26 (5 V) – the relevant pins can be specified by the user. The 3.3 V and the 5 V supply must be switched on simultaneously.

The ADCS has internal power switches for the various individual components (CubeSense, CubeControl, CubeStar, and the CubeWheels), all of which are controlled by CubeComputer. The switch states are available as telemetry (TLM) and can also be toggled using a telecommand (TC).

If CubeComputer is used as the satellite's main OBC (alongside its function as ADCS OBC) it must be supplied with 3.3 V on H2-27,28 (the "always on" 3.3 V).

Currents and voltages are measured at various points on the CubeADCS unit and the measurements can be obtained using TLM requests. The ADCS will also use these measurements to switch off components in case of a latch-up.

The battery bus (pins H2-45 and H2-46) supplies power to the CubeWheels and the magnetometer boom deployment circuitry.

3.2.2 Battery bus voltage

The CubeWheels have a turn-on threshold of 6.5 V, but lower voltages are tolerated without damage to the units. The maximum tested bus voltage is 16.0 V.

3.3 Communication

3.3.1 I²C

The CubeADCS unit can communicate with other satellite subsystems using the System I²C on the PC104 bus (H1-41,43). If CubeComputer is the main OBC of the satellite, it will act as a master on the system I²C bus. Conversely, it will act as a slave on the system I²C bus if it is only the ADCS OBC, responding to commands and telemetry requests (as detailed in the CubeADCS User Manual). CubeComputer can be populated with or without pull-up resistors on the system I²C bus.

3.3.2 UART

The CubeADCS bundle has two UART buses (on CubeComputer), designated UART 1 and UART 2, which can be used to interface with the bundle. Both buses can be connected to the PC104 header and UART 2 is also accessible from the piggyback header on CubeComputer.

The CubeSupport computer software interfaces with CubeComputer via UART 1, which is also accessible via a 3-way 2.54 mm pitch square post female header. The required UART-to-USB cable is supplied with the CubeADCS unit.

3.3.3 CAN

The optional CAN interface on CubeComputer is only available on the main PC104 header (H1-1,3). The termination resistor of the CAN bus is configurable.

3.4 Logging

The CubeADCS bundle has the ability to log data automatically and save it to the SD memory card on CubeComputer. CubeComputer is supplied with an industrial grade 1 GB SD card. For more detail regarding the logging, please consult the CubeADCS User Manual.

3.5 Peripheral components

As shown in Figure 1, the magnetometers and coarse sun sensors interface with CubeControl via wired connections. The three CubeWheels and CubeStar interface with CubeComputer via the CubeConnect interface board in the standard configuration. All of the connections above are polarised for ease of use and the harness lengths are all configurable.

4. Mechanical Interface

4.1 PC104 component stack

Most CubeSpace components make use of the standard “self-stacking” PC104 CubeSat mechanical interface. The dimensions of the stacked components and mounting locations are shown in Figure 5. The CubeADCS stack includes spacers to support the inter-PCB spacing.

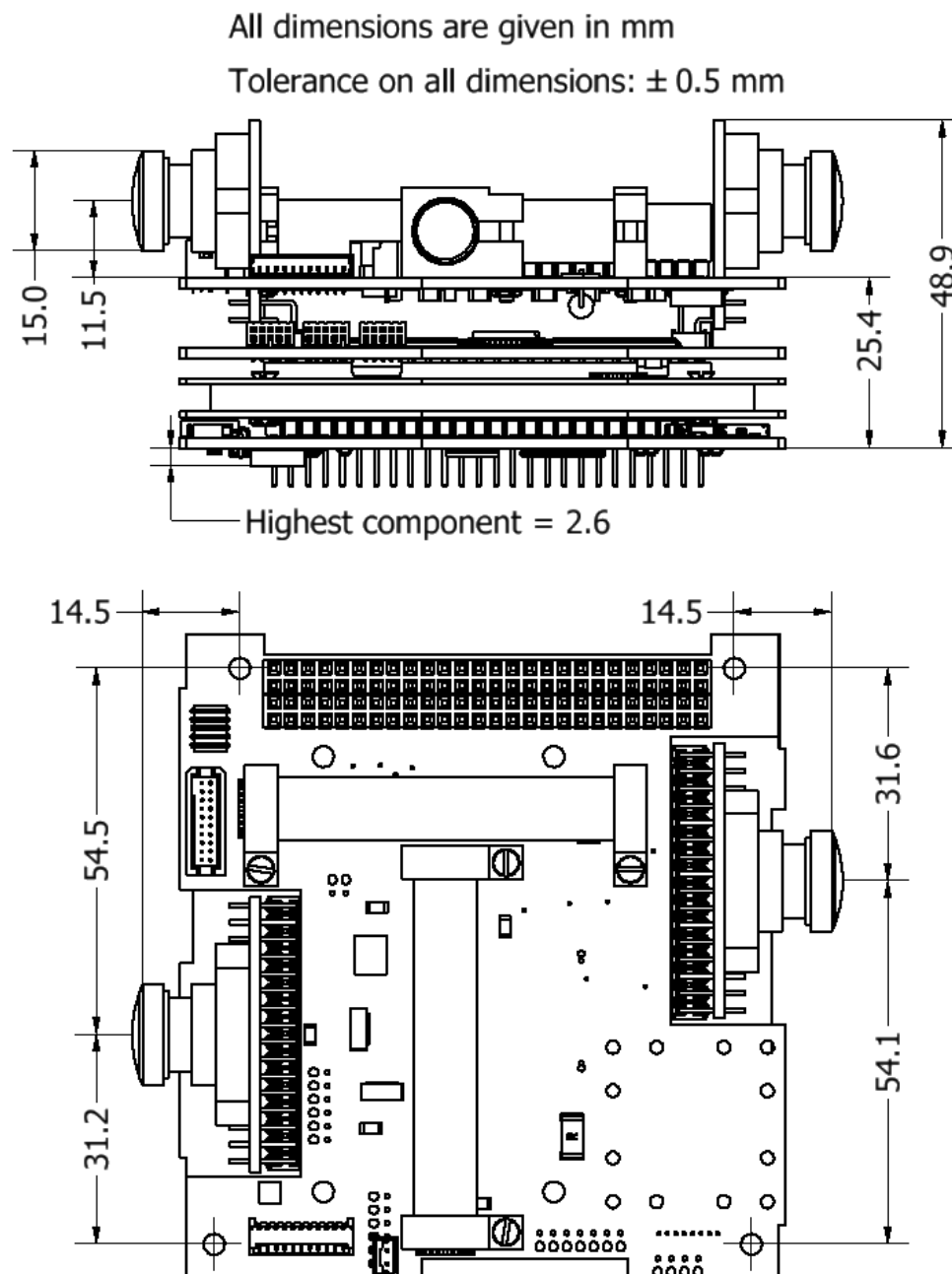


Figure 5 – Dimensions of the PC104 stack of the CubeADCS 3-Axis unit

4.2 Reaction wheels

The mounting of the three CubeWheel reaction wheels will depend on the size of the CubeWheels. In the case of Small CubeWheels, the wheels can be (1) mounted on CubeConnect and integrated in the main PC104 stack, (2) mounted on CubeConnect and supplied on a separate PC104 stack, or (3) supplied loose, i.e. the user must mount the wheels.

4.2.1 Small CubeWheel

The Small CubeWheel unit has two sets of four M2 mounting holes on two different facets, as illustrated in Figure 6. The outer dimensions of the Small CubeWheel are 28 mm x 28 mm x 26.1 mm. Note that an additional 1.5 mm is required on one side of the CubeWheel for internal harnesses. The M2 mounting holes will be fitted with A2 (304) stainless steel heli-coils to provide a stronger and more durable thread.

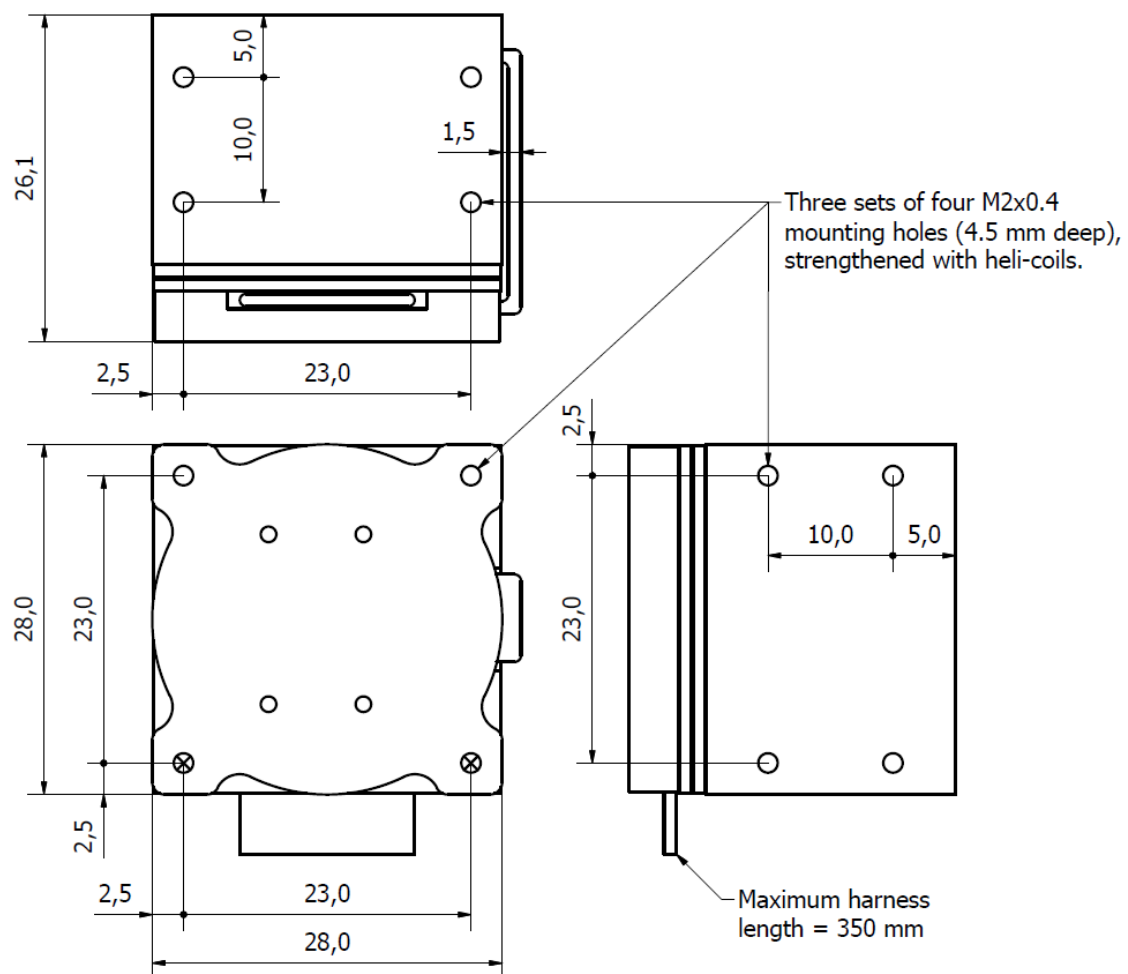


Figure 6 – Small CubeWheel mechanical interface

4.2.2 Medium CubeWheel

The Medium CubeWheel unit has two sets of four M3 mounting holes on two different facets, as illustrated in Figure 7. The outer dimensions of the Medium CubeWheel are 46 mm x 46 mm x 31.5 mm. Note that an additional 1.5 mm is required on one side of the CubeWheel for internal harnesses. The M3 mounting holes will be fitted with A2 (304) stainless steel heli-coils to provide a stronger and more durable thread.

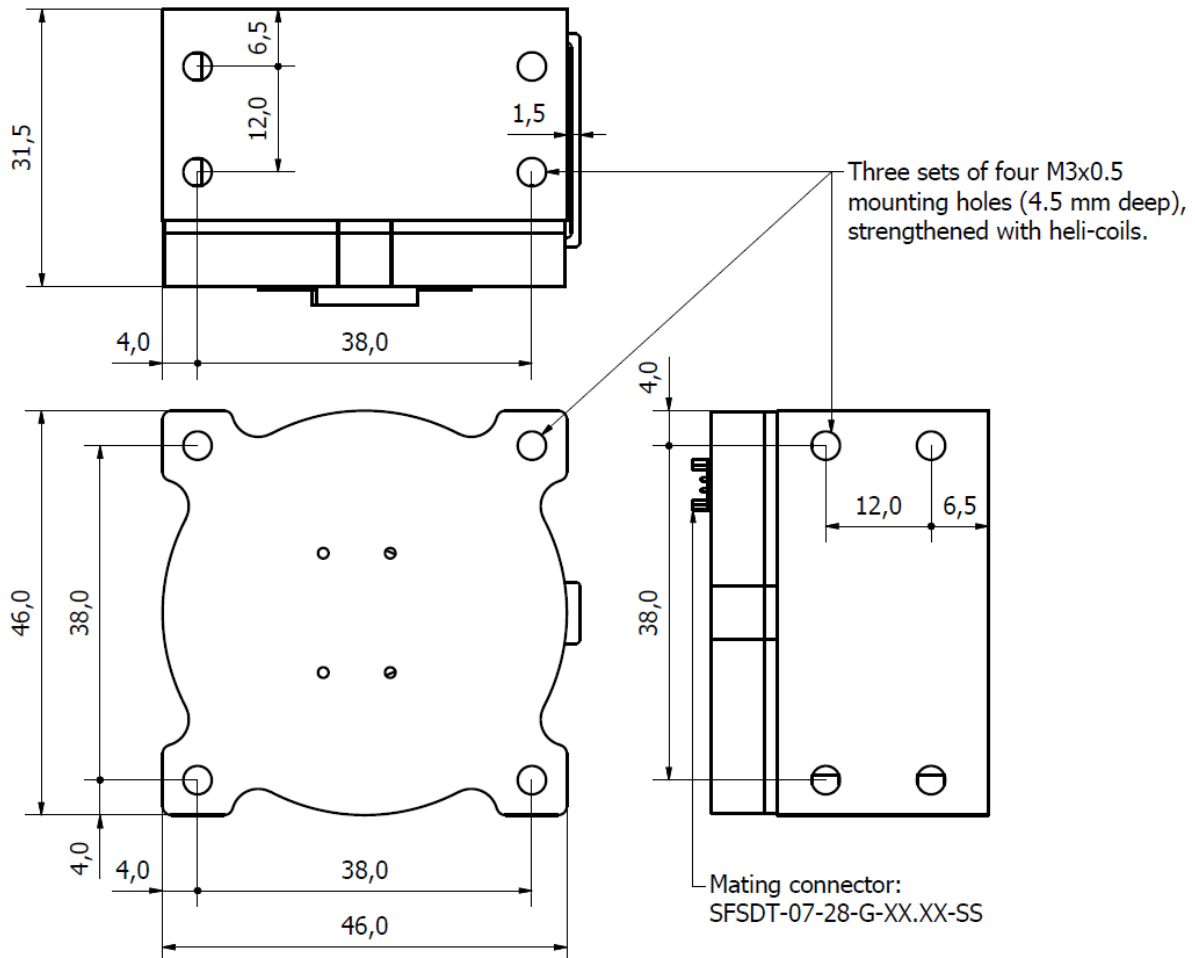


Figure 7 – Medium CubeWheel mechanical interface

4.2.3 Large CubeWheel

The Large CubeWheel unit has two sets of four M3 mounting holes on two different facets, as illustrated in Figure 8. The outer dimensions of the Large CubeWheel are 57 mm x 57 mm x 31.5 mm. Note that an additional 1.5 mm is required on one side of the CubeWheel for internal harnesses. The M3 mounting holes will be fitted with A2 (304) stainless steel heli-coils to provide a stronger and more durable thread.

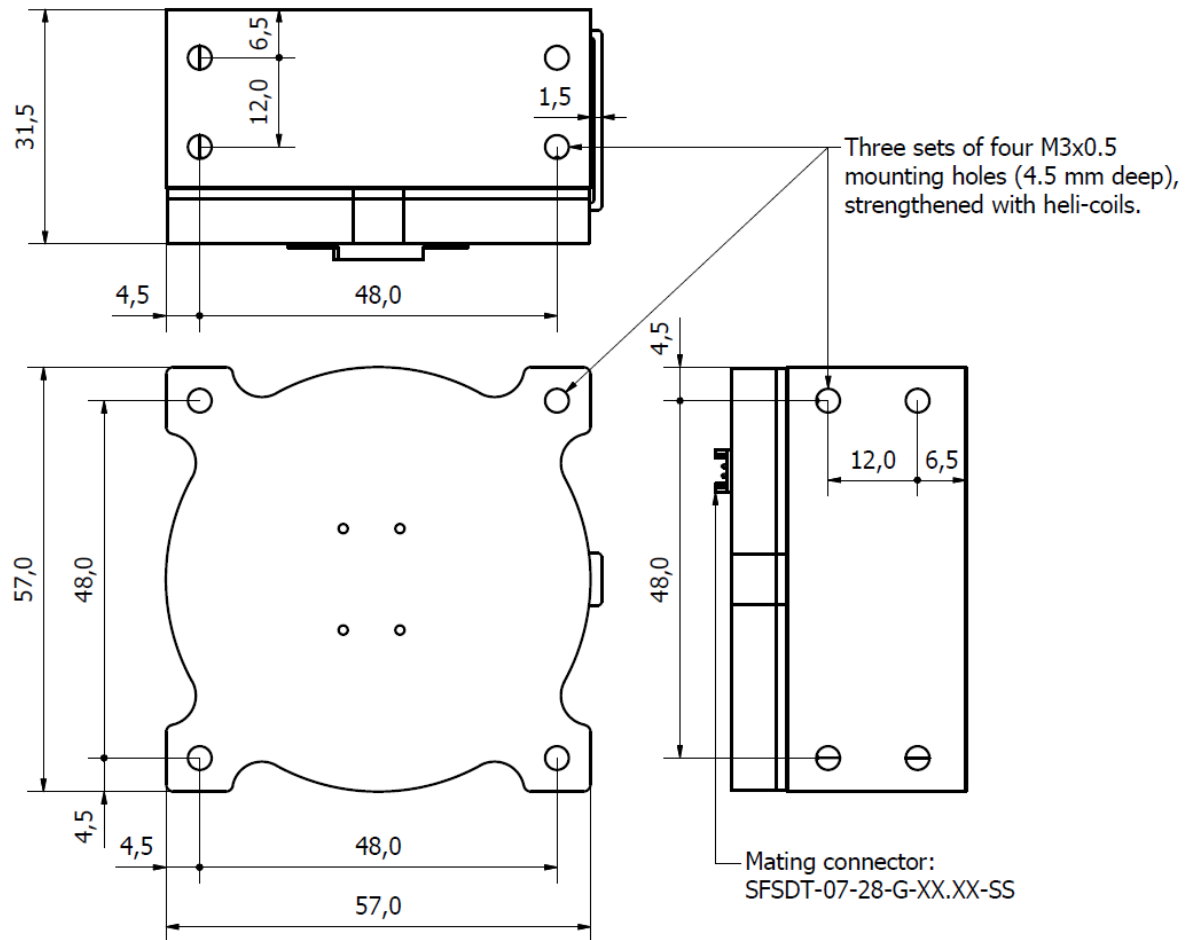


Figure 8 – Large CubeWheel mechanical interface

4.3 Magnetometers

4.3.1 External deployable magnetometer

The primary magnetometer is housed on a small PCB, enclosed by an aluminium housing. The bare magnetometer housing measures 16.8 mm x 17.2 mm x 6.7 mm (as shown in Figure 9).

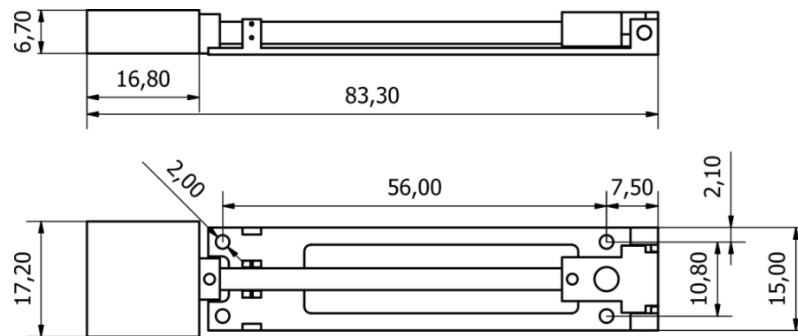


Figure 9 – Deployable magnetometer dimensions and mounting locations

The magnetometer is supplied with a deployable boom by default. The boom separates the magnetometer 8 cm from the satellite body, limiting the effect of electromagnetic interference and lowering the measurement noise. Although the magnetometer can be supplied without the deployment mechanism, it is strongly recommended that the magnetometer should be deployable. The main reason for the aforementioned is the magnetometer's important role during detumbling.

The deployable boom consists of a rod with the magnetometer in its enclosure at one end and a mounting bracket with a spring on the other, as illustrated in Figure 10. The bracket should be mounted on the outside of the satellite structure. The magnetometer will be supplied in the stowed position; deployment can be done in-flight via the appropriate TC.

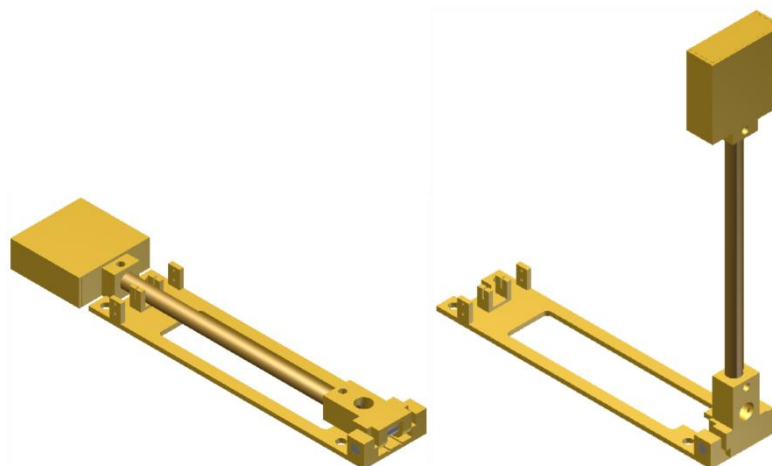


Figure 10 – Deployable magnetometer stowed (left) and deployed (right)

The primary magnetometer and the boom deployment actuation are connected to CubeControl using a single Omnetics Nano Circular 11-way in-line harness. The length between the CubeControl and the in-line connection, as well as the length between the magnetometer and the inline connection, can be configured to the user's requirements. The maximum total length of the harness is 500 mm. The in-line Omnetics connector has a diameter of 3.9 mm.

4.3.2 Redundant magnetometer

The CubeADCS unit offers an interface to an optional redundant magnetometer. The magnetometer is housed in the same enclosure as the primary magnetometer, but it is not deployable.

The redundant magnetometer is connected to the CubeControl module via a 6-way harness (not in-line) and the maximum harness length is 400 mm.

4.4 Coarse sun sensors

The CubeADCS unit can interface with up to 10 coarse sun sensor (CSS) photodiodes, each mounted on a small PCB. The dimensions of each CSS PCB are 3.8 mm x 10.8 mm and the height is 1.7 mm. The CSSs do not have mounting holes – they should be attached to the satellite body using epoxy. At least 6 photodiodes, each on a different facet of the satellite, are required to obtain a coarse sun vector throughout the entire sunlit part of the orbit. The 4 additional sensors can be used for redundancy or in cases where sensors are shadowed or obstructed.

The 10 CSSs are each connected to CubeControl by a Molex PicoBlade 2-way in-line connector to ease the satellite integration process. The ten 2-way harnesses are terminated in a single 20-way female connector, of which the male connector is populated on CubeControl. Figure 11 illustrates the above-mentioned wiring of the CSS units.

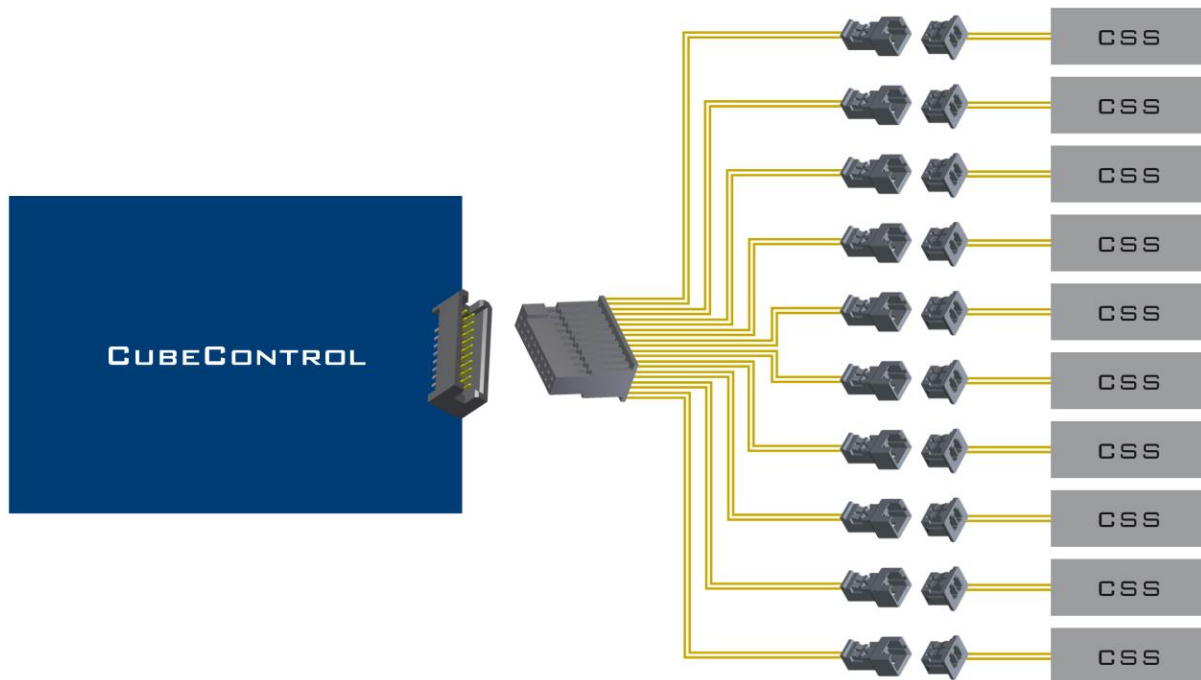


Figure 11 – Coarse sun sensor wiring diagram

4.5 Mass

The approximate masses of the CubeADCS 3-Axis components are shown in Table 2.

Table 2 – Mass of CubeADCS components

Component	Mass (g)	Notes
PC104 components		
CubeControl V2.2	97	Including 2 CubeTorquer rods
CubeComputer V4.1	50	
CubeSense V2.2	79	Including 2 cameras
CubeCoil	46	
Spacers and epoxy	< 10	Depends on configuration
Peripheral components		
3 x Small CubeWheel	195	Including harnesses
3 x Medium CubeWheel	< 480	Including harnesses
3 x Large CubeWheel	< 720	Including harnesses
CubeConnect	46	
PC104 spacers for CubeConnect	28	Only applicable if Small CubeWheels are integrated in bundle
CubeStar	66	Including harness
Deployable magnetometer	15	Including harness and deployment mechanism
Redundant magnetometer	7	Including harness
10 x CSS	15	Including harnesses

5. Software Interface

5.1 UART

The standard specifications of the CubeADCS 3-Axis UART interfaces are given in Table 3. These specifications are however customisable if CubeComputer is the main OBC of the satellite.

Table 3 – Standard UART specifications for CubeComputer

Parameter	Value
Baud rate	115200
Data bits	8
Parity	None
Stop bits	1

The CubeADCS UART protocol includes two characters before and two characters after the TLM or TC packet. More information regarding the UART protocol can be found in the CubeADCS User Manual.

5.2 I²C

If CubeComputer is not used as the main OBC of the satellite (therefore it is only the ADCS OBC), it acts as a slave on the system I²C bus with 7-bit addressing. The default 8-bit write and read addresses of CubeComputer are shown in Table 4.

Table 4 – CubeComputer (slave node) I²C addresses

Function	I ² C address
Write	0x24
Read	0x25

More details regarding the communications interface (including protocols for TCs, TLM requests, and file downloading) can be found in the CubeADCS User Manual.

5.3 CAN

The CubeADCS unit can act as a slave or a master on a 1 Mbps CAN bus.

The combination of a CAN transceiver and a CAN controller module on CubeComputer allows CubeADCS unit to interface at CAN bus voltage levels of 3.3 V or 5 V. A termination resistor between the CANH and CANL lines can be populated if required by the user.

More information regarding the CAN protocol can be found in the CubeADCS User Manual.

5.4 Telecommands and telemetries

5.4.1 Telecommands

Table 5 lists the telecommands for commanding a CubeADCS unit. Note that the list of TCs is compatible with any CubeADCS bundle (Magnetic, Y-Momentum, or 3-Axis) and several TCs will therefore not be relevant for the CubeADCS 3-Axis unit. The identification numbers and structures of the TCs are available once the user has purchased a CubeADCS unit.

Table 5 - List of telecommands

Name	Description	Length
Reset	Perform a reset	1
Current Unix Time	Current Unix Time	6
Cache enabled state	Cache enabled state	1
Reset Log Pointer	Reset pointer to log buffer (from where LastLogEvent TLM is returned)	0
Advance Log Pointer	Advance pointer to log buffer (from where LastLogEvent TLM is returned)	0
Reset Boot Registers	Reset Boot Registers	0
Deploy Magnetometer Boom	Deploy magnetometer boom	1
ADCS Run Mode	Set ADCS enabled state & control loop behaviour	1
ADCS Power Control	Control power to selected components	3
ADCS Configuration	Current configuration	247
Set Magnetorquer Configuration	Set magnetorquer configuration parameters	3
Set Wheel Configuration	Set wheel configuration parameters	4
Set Rate Gyro Configuration	Set rate gyro configuration parameters	3
Set CSS Alignment Configuration	Set photodiode pointing directions	10
Set CSS Scale Factor Configuration	Set photodiode scale factors	11
Set Sun Sensor Configuration	Set sun sensor configuration parameters	14
Set Nadir Sensor Configuration	Set nadir sensor configuration parameters	14
Set Nadir Sensor Mask Configuration 1	Set Nadir Sensor Mask Configuration 1	8
Set Nadir Sensor Mask Configuration 2	Set Nadir Sensor Mask Configuration 2	8
Set Nadir Sensor Mask Configuration 3	Set Nadir Sensor Mask Configuration 3	8
Set Nadir Sensor Mask Configuration 4	Set Nadir Sensor Mask Configuration 4	8
Set Nadir Sensor Mask Configuration 5	Set Nadir Sensor Mask Configuration 5	8
Set Magnetometer Mounting Configuration	Set magnetometer mounting configuration parameters	6
Set Magnetometer Offset and Scaling Configuration	Set Magnetometer Offset and Scaling Configuration	12

Set Magnetometer Sensitivity Configuration	Set Magnetometer Sensitivity Configuration	12
Set Rate Sensor Configuration	Set Rate Sensor Offsets	7
Set Star Tracker Configuration	Set configurations of CubeStar	19
Set Detumbling Control Parameters	Set controller gains and reference values for Detumbling control mode	10
Set Y-Wheel Control Parameters	Set controller gains and reference value for Y-wheel control mode	20
Set Reaction Wheel Control Parameters	Set controller gains and reference value for reaction wheel control mode	8
Set Moments of Inertia	Set satellite moments of inertia	12
Set Products of Inertia	Set satellite products of inertia	12
Set Estimation Parameters 1	Set estimation noise covariance and sensor mask 1	16
Set Estimation Parameters 2	Set estimation noise covariance and sensor mask 2	14
Set SGP4 Orbit Parameters	Set SGP4 Orbit Parameters	64
Set SGP4 Orbit Inclination	Set SGP4 Orbit Inclination	8
Set SGP4 Orbit Eccentricity	Set SGP4 Orbit Eccentricity	8
Set SGP4 Orbit RAAN	Set SGP4 Orbit RAAN	8
Set SGP4 Orbit Argument of Perigee	Set SGP4 Orbit Argument of Perigee	8
Set SGP4 Orbit B-Star Drag term	Set SGP4 Orbit B-Star Drag term	8
Set SGP4 Orbit Mean Motion	Set SGP4 Orbit Mean Motion	8
Set SGP4 Orbit Mean Anomaly	Set SGP4 Orbit Mean Anomaly	8
Set SGP4 Orbit Epoch	Set SGP4 Orbit Epoch	8
Save Configuration	Save current configuration to flash memory	0
Save Orbit Parameters	Save current orbit parameters to flash memory	0
Set Start-up Mode	Select whether the ACP should automatically start Detumbling control at start-up	1
Advance File List Read Pointer	Advance File List Read Pointer	0
Initiate Download Burst	Initiate Download Burst	2
Erase File	Erase File	3
File Upload Packet	File Upload Packet	22
Finalize Upload Block	Finalize Uploaded File Block	7
Hole Map 1	File Upload Hole Map 1	32
Hole Map 2	File Upload Hole Map 2	32
Hole Map 3	File Upload Hole Map 3	32
Hole Map 4	File Upload Hole Map 4	32
Hole Map 5	File Upload Hole Map 5	32
Hole Map 6	File Upload Hole Map 6	32
Hole Map 7	File Upload Hole Map 7	32
Hole Map 8	File Upload Hole Map 8	32
Initiate File Upload	Initiate File Upload	2
Load File Download Block	Fill download buffer with file contents	8
Reset File List Read Pointer	Reset File List Read Pointer	0
Reset Upload Block	Reset HoleMap for Upload Block	0
Set Boot Index	Select which program to boot	1
Copy Program to Internal Flash	Copy Program to Internal Flash	2
Read Program Information	Read Program CRC, length	1

Run Selected Program	Run Selected Program	0
SD Log1 Configuration	Log selection and period for LOG1	13
SD Log2 Configuration	Log selection and period for LOG2	13
UART Log Configuration	Log selection and period for UART (unsolicited TLM)	12
Clear Errors	Clear Latched Error Flags	1
Set Commanded Attitude Angles	Set commanded attitude angles	6
Set Attitude Control Mode	Set attitude control mode	4
Set Attitude Estimation Mode	Set attitude estimation mode	1
Save Image	Save and capture image from one of CubeSense cameras or CubeStar camera to SD card	2
Set Magnetorquer Output	Set magnetorquer output (only valid if Control Mode is None)	6
Trigger ADCS Loop	Trigger ADCS to perform one iteration of the control loop (only valid when ADCS Run Mode is Triggered)	0
Trigger ADCS Loop with Simulated Sensor Data	Trigger ADCS to perform one iteration of the control loop (only valid when ADCS Run Mode is Triggered)	90
Set Wheel Speed	Set wheel speed (only valid if Control Mode is None)	6

5.4.2 Telemetry

Table 6 lists the telemetry frames that can be requested from a CubeADCS unit. Note that the list of TLMs is compatible with any CubeADCS bundle (Magnetic, Y-Momentum, or 3-Axis) and several TLMs will therefore not be relevant for the CubeADCS 3-Axis unit. The identification numbers and structures of the TLM frames are available once the user has purchased a CubeADCS unit.

Table 6 - List of telemetry frames

Name	Description	Length
Identification	Identification information for this node	8
Boot And Running Program Status	Boot And Running Program Status	6
Boot Index and Status	Current selected boot index and status of last boot	2
Cache enabled state	Cache enabled state	1
Current Unix Time	Current Unix Time	6
Last Logged Event	Last Logged Event (relative to pointer - adjusted via Advance and Reset TCs (3 & 4))	6
SRAM Latchup counters	SRAM Latchup counters	6
EDAC Error Counters	EDAC Error Counters	6
Communication Status	Communication status - includes command and telemetry counters and error flags	6
Current ADCS State	Current state of the Attitude Control Processor	6
Estimated Attitude Angles	Estimated attitude angles	6
Estimated Quaternion	Estimated quaternion set	6
Estimated Angular Rates	Estimated angular rates relative to orbit reference frame	6
Satellite Position (ECI)	Satellite position in ECI frame	6
Satellite Velocity (ECI)	Satellite velocity in ECI frame	6
Satellite Position (LLH)	Satellite position in WGS-84 coordinate frame	6
ECEF Position	Satellite position in ECEF coordinates	6
Magnetic Field Vector	Measured magnetic field vector	6
Coarse Sun Vector	Measured coarse sun vector	6
Fine Sun Vector	Measured fine sun vector	6
Nadir Vector	Measured nadir vector	6
Rate Sensor Rates	Rate sensor measurements	6
Wheel Speed	Wheel speed measurement	6
Star 1 Body Vector	Star 1 Body Vector	6
Star 1 Inertial Vector	Star 1 Inertial Vector	6
Star 2 Body Vector	Star 2 Body Vector	6
Star 2 Inertial Vector	Star 2 Inertial Vector	6
Star 3 Body Vector	Star 3 Body Vector	6
Star 3 Inertial Vector	Star 3 Inertial Vector	6
Magnetorquer Command	Magnetorquer commands	6
Wheel Speed Commands	Wheel speed commands	6
IGRF Modelled Magnetic Field Vector	IGRF modelled magnetic field vector (orbit frame referenced)	6

Modelled Sun Vector	Modelled sun vector (orbit frame referenced)	6
Estimated Gyro Bias	Estimated rate sensor bias	6
Estimation Innovation Vector	Estimation innovation vector	6
Quaternion Error Vector	Quaternion error vector	6
Quaternion Covariance	Quaternion covariance	6
Angular Rate Covariance	Angular rate covariance	6
Raw Nadir Sensor	Nadir sensor capture and detection result	6
Raw Sun Sensor	Sun sensor capture and detection result	6
Raw CSS 1 to 6	Raw CSS measurements 1 to 6	6
Raw CSS 7 to 10	Raw CSS measurements 7 to 10	6
Raw Magnetometer	Raw magnetometer measurements	6
Star Performance	Performance parameters of star measurement	6
Star Magnitude	Instrument magnitude of identified stars	6
Star 1 Raw Data	Catalogue index and detected coordinates for star 1	6
Star 2 Raw Data	Catalogue index and detected coordinates for star 2	6
Star 3 Raw Data	Catalogue index and detected coordinates for star 3	6
Star Timing	Timing information of star measurement	6
Raw GPS Status	Raw GPS status	6
Raw GPS Time	Raw GPS time	6
Raw GPS X	Raw GPS X position and velocity (ECI referenced)	6
Raw GPS Y	Raw GPS Y position and velocity (ECI referenced)	6
Raw GPS Z	Raw GPS Z position and velocity (ECI referenced)	6
CubeSense Current Measurements	CubeSense current measurements	6
CubeControl Current Measurements	CubeControl current measurements	6
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Raw Sensor Measurements	Raw sensor measurements	28
Raw Star Tracker	Raw Star Tracker Measurement	36
Power and Temperature Measurements	Power and temperature measurements	32
ADCS Measurements	Calibrated sensor measurements	72
Actuator Commands	Actuator commands	12
Adcs Execution Times	Returns information about execution times of ACP functions	8
ADCS Power Control	Control power to selected components	3
ADCS Configuration	Current configuration	247
Set SGP4 Orbit Parameters	Set SGP4 Orbit Parameters	64
Set Commanded Attitude Angles	Set commanded attitude angles	6

Program Information	Program information including file size and CRC	8
Copy To Internal Flash Progress	Progress of copy to internal flash operation	1
Status of Image Capture and Save Operation	Status of Image Capture and Save Operation	2
SD Log1 Configuration	Log selection and period for LOG1	13
SD Log2 Configuration	Log selection and period for LOG2	13
UART Log Configuration	Log selection and period for UART (unsolicited TLM)	12
Download Block Ready	Status about download block preparation	5
File Information	File Information	12
Initialize Upload Complete	Initialize Upload Complete	1
Upload Block Complete	Finalize Upload Block Complete	1
Block Checksum	File upload Block CRC16 Checksum	2
Hole Map 1	File Upload Hole Map 1	32
Hole Map 2	File Upload Hole Map 2	32
Hole Map 3	File Upload Hole Map 3	32
Hole Map 4	File Upload Hole Map 4	32
Hole Map 5	File Upload Hole Map 5	32
Hole Map 6	File Upload Hole Map 6	32
Hole Map 7	File Upload Hole Map 7	32
Hole Map 8	File Upload Hole Map 8	32
ACP Execution State	Returns information about the ACP loop	3

6. Specifications

6.1 CubeADCS specifications

Table 7 – CubeADCS 3-Axis specifications

Specification	Value	Notes
Electrical		
3.3 V supply	3.23 V – 3.40 V	
5 V supply	4.85 V – 5.15 V	
V _{battery}	6.5 V – 16.0 V	
Physical		
Mass		
CubeADCS PC104 stack	282 g	See Section 4.5 for breakdown
Peripherals	372 g < 630 g < 870 g	Incl. Small wheels, CubeStar Incl. Medium wheels, CubeStar Incl. Large wheels, CubeStar
Dimensions		
PC104 stack	90 x 96 x 52 mm 90 x 96 x (68 – 80)** mm **Depending on configuration	Without CubeWheels Small CubeWheels integrated
Small CubeWheel	28 x 28 x 26.1 mm	
Medium CubeWheel*	46 x 46 x 31.5 mm	
Large CubeWheel*	57 x 57 x 31.5 mm	
CubeStar*	35 x 50 x 57 mm	Including lens
CSS	3.8 x 10.8 x 1.7 mm	Each photodiode PCB
Deployable magnetometer	16.8 x 83.3 x 6.7 mm	Stowed
Redundant magnetometer*	19 x 19 x 6.7 mm	
Outgassing		
Total Mass Loss (TML)	< 1.0 %	
Collected Volatile Condensable Material (CVCM)	< 0.1 %	
Vibration		
Sine sweep	5 - 100 Hz @ 2.5 g 100 - 125 Hz @ 1.25 g	Sweep rate: 2 oct/min Direction: X, Y, Z
Random vibration	20 Hz @ 0.01125 g ² /Hz 130 Hz @ 0.05625 g ² /Hz 800 Hz @ 0.05625 g ² /Hz 2000 Hz @ 0.015 g ² /Hz	Duration: 120 s RMS acceleration: 8.03 g Direction: X, Y, Z
Thermal		
Operating temperature range	-10°C to +60°C	
Storage temperature range	-10°C to +60°C	

* Optional extra on CubeADCS 3-Axis

6.2 Sensor specifications

Table 8 – Sensor specifications

Specification	Value	Notes
Sun and nadir sensor		
Sun sensor measurement accuracy	$< 0.2^\circ$	Entire FOV, $3\text{-}\sigma$
Nadir sensor measurement accuracy	$< 0.2^\circ$	Entire Earth visible in FOV, $1\text{-}\sigma$
Field of view (both cameras)	$150^\circ - 200^\circ$	
Update rate	1 Hz	
Image size	1 MB	1024 x 1024 pixels
Image format	8-bit grayscale	
Coarse sun sensors		
Measurement accuracy	$< 10^\circ$	1σ
Update rate	1 Hz	
Magnetometers		
Measurement noise	$< 50 \text{ nT}$	1σ (per X/Y/Z channel)
Update rate	1 Hz	Averaged output
Rate sensors		
Measurement noise	$< 0.015^\circ/\text{s}$	1σ
Update rate	1 Hz	Averaged output
Angular Random Walk	$0.28^\circ/\sqrt{h}$	
Bias instability	$24^\circ/\text{h}$	

6.3 Actuator specifications

Table 9 – Actuator specifications

Specification	Value	Notes
Magnetorquers		
Maximum magnetic dipole (rods)	0.2 Am ²	Maximum on-time of 800 ms
Maximum magnetic dipole (coil)	0.12 Am ²	Maximum on-time of 800 ms
On-time command resolution	10 ms	For a 1 Hz control period
Small CubeWheel		
Maximum momentum storage	1.7 mNms	
Maximum wheel speed	± 8000 rpm	
Maximum torque	0.15 mNm	
Medium CubeWheel		
Maximum momentum storage	10.0 mNms	
Maximum wheel speed	± 6000 rpm	
Maximum torque	1.0 mNm	
Large CubeWheel		
Maximum momentum storage	30.0 mNms	
Maximum wheel speed	± 6000 rpm	
Maximum torque	2.3 mNm	

6.4 Power consumption

Table 10 – Power consumption

Component	Average current (mA)			Total average power (mW)
	3.3 V	5 V	V _{battery} *	
CubeComputer	60	-	-	198
CubeSense	10	14	-	103
CubeControl (torquers off)	75	-	-	248
CubeControl (torquers full on)	75	230	-	1398
Small CubeWheel (@ 2000 rpm)	21	-	7	125
Medium CubeWheel (@ 2000 rpm)	21	-	TBD	< 260
Large CubeWheel (@ 2000 rpm)	21	-	TBD	< 300

* V_{battery} = 8.0 V

7. Handling

Anti-static

The bundle contains a variety of static sensitive devices. The appropriate electrostatic discharge (ESD) protection measures must therefore be implemented. The unit must never be handled without proper grounding.

Cleanliness

It is recommended that the CubeADCS unit be handled in a clean environment. A clean room of ISO class 8 or better or an appropriate laminar flow workbench is recommended.

Moisture

The unit should be kept free of moisture or liquids, as these could have corrosive effects on the electronics and electronic joints which may lead to degradation and loss of reliability of the circuits.

Shock

The unit must be handled with care and dropping or bumping the unit should be avoided completely.

Camera lens cleanliness

The camera lenses should be kept clean and free of any dirt that may obstruct the images captured by the camera. Dust should be removed with a microfibre cloth. If required, the lens may be cleaned using ethanol and appropriate lens cleaning equipment, but unnecessary cleaning of the lens should be avoided.

Camera lens structural integrity

The camera sensors are aligned to be parallel with the CubeSense PCB. This is important since misalignment of the cameras influence the performance of the system. External forces on the camera modules should therefore be avoided entirely.

Camera lens covers

The sun and nadir optics are fitted with dust caps which should be removed before flight.

Reaction wheels

The aluminium housings of the reaction wheels should NOT be tampered with. Tampering with the housings may damage the wheels. No attempt should be made to loosen or remove the fasteners that secure the housings.

8. Document Version History

Version	Author(s)	Pages	Date	Description of change
0.1	GJVV	ALL	08/01/2016	First draft
0.2	GJVV	8, 10, 14	19/01/2016	Minor corrections
0.3	GJVV	7	26/01/2016	Removed erroneous reference to Y-Momentum
0.4	GJVV	ALL	04/02/2016	Various minor updates and corrections
0.5	GJVV	Title	22/02/2016	Changed image on front page
0.6	GJVV	7	22/02/2016	Removed CC I ² C buffer power, minor edits
0.7	JG	4, 8, 10	22/06/2016	Minor corrections
1.0	GJVV	ALL	27/06/2016	Several updates
1.1	MK	ALL	29/06/2016	Review and corrections
1.2	GJVV	ALL	22/07/2016	Minor corrections, including new CubeWheel drawings
1.3	GJVV	Title, 11	04/08/2016	Updated title image and CAD drawing of stack